26-ES-EE101 — Basic Electrical and Electronics Engineering

Course Code	26-ES-EE101	**Category**	Engineering Science Course
Course Title	Basic Electrical and Electronics Engineering	**L – T – P**	3 – 1 – 0
Credits	4	**Contact Hours**	44
Semester	First	**Assessment**	Theory: CA 30 + ESE 70

Prerequisites

A four-credit course (3-1-0), the tutorial hour serving the problem-working that a combined electrical-and-electronics paper needs. Scoped at about forty-four hours. Assumed:

- Ohm's and Kirchhoff's laws and simple resistive circuits (CBSE XII Unit II; WBCHSE).
- Alternating current and the LCR circuit in outline (CBSE XII Unit IV).
- The p–n junction diode and the elements of the transistor and logic gates (CBSE XII Units IX–XI).
- The calculus of Mathematics-I, taken concurrently.

Course Objectives

- To establish the analysis of direct- and alternating-current circuits by the network theorems.
- To introduce electrical machines and their engineering use.
- To establish the physics and characteristics of the semiconductor devices — diode, transistor, field-effect transistor and operational amplifier.
- To ground the student in electrical safety and the Indian wiring practice.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Apply electrical safety and wiring practice and analyse direct-current circuits using the network theorems.	L3 – Apply
CO2	Analyse single-phase and balanced three-phase alternating-current circuits and compute power and power factor.	L3 – Apply
CO3	Explain the construction and characteristics of transformers and rotating machines and their engineering use.	L2 – Understand
CO4	Explain the physics and characteristics of the semiconductor diode and its applications in rectification and regulation.	L3 – Apply
CO5	Calculate the voltage gain of the operational amplifier in the inverting and non-inverting configuration and study its applications.	L3 – Apply

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	2	1	1	1	–	–	1
CO2	3	3	2	2	2	–	–	–	–	–	1
CO3	3	2	2	1	2	1	1	–	–	–	1
CO4	3	3	2	2	2	–	–	–	–	–	2
CO5	3	3	3	2	3	–	–	1	–	–	2

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours
I	Basics of Electrical Wiring and Safety	Electrical safety — the effect of current on the body; earthing; the residual-current device; the Switch-Fuse Unit (SFU), MCB, ELCB and MCCB; types of wires and cables; the elements of the Indian wiring practice; types of batteries and their important characteristics.	3
II	Direct-Current Circuits	Circuit elements and their voltage–current relations; independent and dependent sources; Kirchhoff's voltage and current laws; mesh and nodal analysis; star–delta transformation; superposition, Thevenin, Norton and maximum-power-transfer theorems; transient response of RL and RC circuits in outline.	9
III	Alternating-Current Circuits	Generation of alternating voltage; average and root-mean-square values, form and peak factors; phasor representation; analysis of R, L, C and series and parallel RLC circuits; impedance and admittance; real, reactive and apparent power, power factor and its improvement; series and parallel resonance; balanced three-phase systems, star and delta connection, and three-phase power measurement in outline.	8
IV	Transformers and Rotating Machines	Single-phase transformer — construction, principle, equivalent circuit, losses, efficiency and regulation; the auto-transformer in outline. Direct-current machines — construction, e.m.f. and torque equations, types of excitation, characteristics and starting. Introduction to alternating-current motors (three-phase and single-phase), with the stepper motor and the brushless direct-current motor and a note on their use in automation.	6
V	Semiconductor Fundamentals and Diode Applications	Energy bands in solids; valence and conduction bands; insulators, semiconductors and conductors; intrinsic and extrinsic semiconductors; current in semiconductors by diffusion and drift; the p–n junction diode under forward and reverse bias and its current–voltage characteristic; breakdown — Zener and avalanche; rectifiers — half-wave and full-wave with filters; the Zener diode as a voltage regulator; the light-emitting diode and the photodiode.	7
VI	Transistors and Operational Amplifiers	Bipolar junction transistor — structure and symbol, operation, modes, the CB, CE and CC characteristics and current-gain expressions, the Early effect, and the hybrid (h-parameter) model in outline. Field-effect transistor — the junction FET: circuit, symbol, drain and transfer characteristics, and the saturation drain current; the MOSFET — construction, operation and drain characteristics. The operational amplifier — the ideal op-amp, open-loop characteristics and parameters; the inverting, non-inverting, summing and difference amplifier.	11

Note on scope and delivery. The course carries four credits as 3-1-0 — three lectures and a tutorial — in recognition that it combines two subjects within one paper. The tutorial hour is for the working of circuit and device problems and is not a fourth lecture. The safety-and-wiring Module I and the semiconductor and operational-amplifier content of Modules V and VI are the parts strengthened at the Board's instance, so that the electronics half is given its due.

Text and Reference Books

- S.K. Sahdev, Basic Electrical Engineering (with Lab Manual), Khanna Publishing House (AICTE Prescribed Textbook)
- Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House.
- D. P. Kothari and I. J. Nagrath, Basic Electrical and Electronics Engineering, McGraw-Hill.
- D. C. Kulshreshtha, Basic Electrical Engineering, McGraw-Hill.
- Boylestad and Nashelsky, Electronic Devices and Circuit Theory, Pearson.
- V. K. Mehta and Rohit Mehta, Principles of Electronics, S. Chand.
- Millman, Halkias and Jit, Electronic Devices and Circuits, McGraw-Hill.

Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Basic Electrical Circuits / Basic Electrical Engineering (IITs), 12 weeks.
- NPTEL — Electronic Devices and Circuits, 12 weeks.
- NPTEL — Analog Electronic Circuits / Operational Amplifiers, 12 weeks.
- Virtual Labs (Ministry of Education) — Basic Electrical and Analog Electronics laboratories.

Question-paper note. The paper is a problem-and-reasoning paper. At least one Group C question requires network analysis by a stated theorem, one requires an op-amp gain calculation (CO5), and one requires interpretation of a device characteristic. Circuit diagrams are to be drawn to standard symbols.